Smart Parking Availability Prediction using Explainable AI

AI Foundations for Engineers

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1 Abstract

This project presents an explainable AI system for predicting short-term parking availability near a selected area. The model combines live parking-slot status, historical parking logs, nearby-place activity, route context, time features, and environmental signals from Google Maps Platform APIs. The system is designed around core syllabus topics from AI Foundations for Engineers, including rational agents, search algorithms, decision tree learning, Bayesian learning, instance-based learning, neural-network concepts, and responsible AI.

2 Introduction

Finding parking in busy urban areas is a dynamic problem. Parking availability changes according to time, nearby commercial activity, weather, road access, and user behavior. A static parking display cannot reliably answer whether a slot will still be available after the user reaches the area. The goal of this project is to build an AI agent that estimates near-future parking availability and explains its prediction to the user.

3 Problem Statement

The problem is to predict the number of parking slots likely to remain available in a short time horizon, such as 30 minutes, using data from the parking system and external APIs. The model must work even when some API signals are unavailable.

4 Objectives

- Build a working smart parking web application.
- Keep the AI model separated from the user interface.
- Use Google Maps Platform APIs as external signal sources.
- Convert API responses into useful AI features.
- Implement explainable AI concepts from the syllabus.
- Display prediction confidence and responsible AI explanations.

5 System Architecture

The system has three main layers:

User interface Allows users and admins to interact with parking areas, slot-status displays, simulated occupancy, and predictions.

Data layer Stores parking areas, slots, logs, and simulated sensor states.

AI layer Extracts features, runs explainable prediction models, and returns area-level availability estimates with confidence.

```
User Interface
  -> Parking Data
  -> Feature Extraction
  -> Explainable AI Model
  -> Responsible AI Explanation
  -> Prediction Output
```

6 Google API Signal Sources

The project key has 33 Google Maps Platform APIs enabled. The AI design uses relevant APIs as optional signal sources. Not every API should be called for every prediction because that would waste quota and introduce unnecessary noise.

Table 1: Main API signals used for AI prediction

API	Feature	Purpose
Places API	Nearby shops and opening hours	Demand estimation
Weather API	Rain and weather condition	Duration and walking penalty
Geocoding API	Address to coordinates	Location input
Distance Matrix API	Travel time	Search cost
Routes API	Route alternatives	Recommendation cost
Roads API	Road access	Entrance/accessibility score
Time Zone API	Local time	Correct peak-hour features
Air Quality API	AQI	Environmental comfort
Pollen API	Pollen level	Health comfort signal
Elevation API	Slope	Walking/accessibility signal

7 AI Foundations Mapping

7.1 Rational Agent

The smart parking predictor is an intelligent agent. Its sensors are API signals, live slot status, history logs, and user location. Its actions are predicting availability, showing confidence, and explaining risk. Its performance measure is the usefulness and accuracy of the prediction.

7.2 Search Problem

The parking task can be formulated as a search problem. The initial state is the user's current location and current parking conditions. The goal state is selecting a parking area

with a higher probability of available space. Actions include choosing an area, waiting, or rerouting.

$$\begin{aligned} cost = {} & distance + travelTime + occupancyRisk \\ & + demandRisk + weatherPenalty - confidenceBonus \end{aligned} \tag{1}$$

An A-star style heuristic can rank the best parking option using this cost.

7.3 Decision Tree Learning

The decision tree classifies demand into low, moderate, high, or very high risk.

```
IF live occupancy >= 82%
  THEN Very high demand
ELSE IF nearby demand is high AND occupancy is medium
  THEN High demand
ELSE IF historical pressure is high
  THEN High demand
ELSE IF demand or occupancy is moderate
  THEN Moderate demand
ELSE
  THEN Low demand
```

7.4 Bayesian Learning

The Bayesian model estimates parking pressure from evidence:

$$P(highPressure \mid peakHour, nearbyDemand, weather, history) \tag{2}$$

This makes the model suitable for uncertain real-world conditions.

7.5 Instance-Based Learning

Instance-based learning compares the current case with similar historical cases. For example, if the current case is Saturday evening with restaurants nearby and rainy weather, the model can compare it with similar past records and use their average occupancy trend.

7.6 Neural Network Concept

The project uses a single-layer sigmoid score:

$$neuralScore = \frac{1}{1 + e^{-z}} \tag{3}$$

$$z = b + w_1x_1 + w_2x_2 + w_3x_3 + w_4x_4 + w_5x_5 \tag{4}$$

The features include live occupancy, nearby demand, historical pressure, weekend flag, and peak-hour flag.

8 Prediction Model

The final prediction can be represented as an ensemble:

$$\begin{aligned} predictedOccupancy = {} & 0.35(liveOccupancy) + 0.20(nearbyDemand) \\ & + 0.15(historicalPressure) + 0.10(weatherImpact) \\ & + 0.10(routeAccessCost) + 0.05(bayesianPressure) \\ & + 0.05(instancePressure) \end{aligned} \tag{5}$$

The output is converted into predicted available slots, occupancy percentage, risk label, demand label, and confidence. The system does not book, reserve, or allocate a specific parking spot because model prediction is not the same as physical confirmation.

9 Responsible AI

The project follows responsible AI principles by showing confidence and explanation instead of only showing a prediction. The interface should tell users whether live API data or fallback local data was used. It should avoid false certainty and should not store unnecessary precise user-location history.

10 Implementation

The AI model is implemented separately from the website UI in:

```
js/ai/parking-availability-ai.js
```

The website imports the AI model and displays the prediction in the user dashboard.

11 Testing and Evaluation

The project can be evaluated using:

- Syntax checks for JavaScript modules.
- Demo data with simulated sensor updates.
- Comparison between predicted availability and actual later slot state.
- Error metrics such as mean absolute error.
- User-facing checks for explanation clarity.

12 Limitations

- Exact Google Maps "Popular times" data is not available through public APIs.
- API calls should be moved to a backend proxy for production security.
- More real parking logs are needed for stronger model evaluation.
- Predictions are estimates and not guarantees.

13 Future Work

Future improvements include adding Weather API signals, Air Quality signals, Pollen signals, Routes API travel cost, Roads API access scoring, KNN-based historical comparison, backend API security, persistent cloud storage, and model monitoring.

14 Conclusion

This project demonstrates how a smart parking system can use explainable AI to predict parking availability. It connects practical web development with syllabus concepts including intelligent agents, search algorithms, decision tree learning, Bayesian learning, instance-based learning, neural scoring, and responsible AI.

References

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